Scribing STAT 578 (Week of Sep 30)

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Lecture Sep 30

Last lecture: perfect recovery, we had a general result which was that: (same argument, slightly restated, the main thing we needed was some type of concentration inequality $\|e_i^\top E\hat{U}^{-i}\|$ (Abbe et al. assumption)

Theorem 0.1

Suppose

$$\hat{M} = M + E$$

Assume that $\text{var}(E_{ij}) \leq \sigma^2$

$$|E_{ij}| \leq 1 \quad \text{a.s.} \quad \frac{\lambda_r}{\sigma} \geq c\sqrt{rn\log(n)}$$

Then

$$\inf_0 \|\hat{U} - U_0\|_{2,\infty} \leq \frac{c\sigma\sqrt{n\log(n)}}{\lambda_r}\|U\|_{2,\infty}$$

Application 1: 2-Block (abba) SBM We'll apply this to the simple 2-block SBM. Let's assume that we have $\frac{n}{2}$ vertices in each row.

$$A \mapsto \hat{U} \in \mathbb{R}^{n\times 2} \mapsto \hat{U}_{\cdot 2}$$

We'll output $\hat{c}(i) = \text{sign}(\hat{U}_{\cdot 2})$
Take $a = \frac{c_1\log(n)}{n} \quad b = \frac{c_2\log(n)}{n} \quad c_1 > c_2$

$$\frac{c_1 - c_2}{c_2} \leq c\sqrt{n\log(n)} \checkmark$$

$$\frac{(c_1 - c_2)}{\sqrt{c_1}}\sqrt{\frac{\log(n)}{n}} \geq c\frac{\sqrt{\log(n)}}{n} \checkmark$$

Then we get perfect clustering.
We also have a more general theorem for K-Block SBMs.

Theorem 0.2

$$B = \rho_n B_0$$

B_0 is constant in n.

$$|c_k| \asymp \frac{n}{k}, \quad n\rho_n \geq c\log(n)$$

Then K-means applied to rows of $\hat{U}$ perfectly recovers communities with high probability.

A lot of the times, it's the spectral algorithms that can achieve perfect recovery (according to the information theorists).

From random graph theory:

$$\frac{n\rho_n}{\log(n)} \to 0 \implies \text{A is disconnected almost surely}$$

$$n\rho_n \to \infty \implies \text{A has a "giant component"}$$

See Anderson Zhang (likes to find the exact constant for whatever he's doing, so just leave it to him to chase constants which is hard and sometimes helpful) on "Fundamental Limits..." in IEEE Information Theory. Does vanilla spectral clustering (not using any information about likelihood), e.g. running K-means on something like eigenvectors, if you do that procedure there is actually a constant-order difference between what vanilla and optimal is. But that constant order only shows up in the above regime.

Theorem 0.3

$$\hat{c} = c \quad \textbf{up to relabeling with high probability}$$

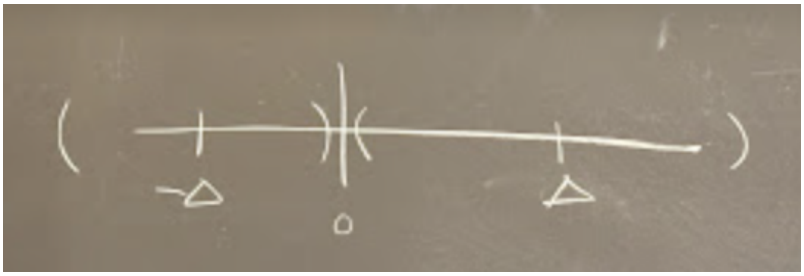
Assumption:

$$a > b,\ a \ll 1,\ \frac{a-b}{b} \le c\sqrt{n\log(n)}$$

$$\frac{a-b}{\sqrt{a}} \ge c_1\sqrt{\frac{\log(n)}{n}}$$

The idea is that somehow the signs of all the $\hat{U}$ are the same from the above theorem. We can show that $\|\hat{U}_{\cdot 2} \pm U_{\cdot 2}\|_\infty$. We'll label the communities as 1, -1. We claim that c and $\hat{c}$ are the same up to relabeling.

$$\|\hat{U}_{\cdot 2} \pm U_{\cdot 2}\|_\infty < \min_i |U_{i2}| = \Delta$$



We know that $U = \frac{1}{\sqrt{n}} \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \\ 1 & -1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}$

Lemma 0.4

$$|U_{i2}| = \frac{1}{\sqrt{n}}$$

$$\lambda_1 = \frac{a+b}{2}n$$

$$\lambda_2 = \frac{a-b}{2}n$$

Theorem $\implies \|\hat{U} - UO^*\|_{2,\infty} \le \frac{c\sqrt{a(1-a)n\log(n)}}{n\frac{a-b}{2}}\|U\|_{2,\infty} \asymp \frac{c\sqrt{a\log(n)}}{n(a-b)}$.

We know from before $O^* = \arg\min_{OO^\top = I_r} \|\hat{U} - UO\|_F$

Claim 0.5

O^* is **close** to a diagonal matrix. Recall that we compute this from the singular values of $\hat{U}^\top U$. Not only is there an Eigengap between the second and third, but the first and second. We can simply quantify

$$O^* - U^\top \hat{U} \lesssim \frac{\sigma^2 n}{\lambda_2}$$

The off-diagonals are exactly $|\hat{U}_{\cdot 1}^\top \hat{U}_{\cdot 2}| = \cos(U_{\cdot 1}, \hat{U}_{\cdot 2})$.

First we can claim that

$$\|U_{\cdot 1} \pm \hat{U}_{\cdot 1}\| \le \frac{c\sigma\sqrt{n}}{\lambda_1 - \lambda_2} = \frac{c\sqrt{a}}{\sqrt{n}b}$$
$$\|U_{\cdot 2} \pm \hat{U}_{\cdot 2}\| \le \frac{c\sigma\sqrt{n}}{\lambda_1 - \lambda_2} = \frac{c\sqrt{a}}{\sqrt{n}b}$$

$$\left\| U^\top \hat{U} - \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \right\| \le 2 \max_{i,j} \left| U_{\cdot i}^\top \hat{U}_{\cdot j} - \mathbf{1}\{i = j\} \right| \le \frac{c\sqrt{a}}{\sqrt{n}b}$$
$$\|U_{\cdot 1}^\top \hat{U}_{\cdot 2}\| \le \|[U_{\cdot 1}, U_\perp]^\top \hat{U}_{\cdot 2}\| = \sin\theta(U_{\cdot 2}, \hat{U}_{\cdot 2})$$
$$\|\hat{U} - UD\|_{2,\infty} \le \|\hat{U} - UO^*\|_{2,\infty} + \|U(O^* - D)\|_{2,\infty} \le \frac{C\sqrt{a\log(n)}}{\sqrt{n}(a-b)} + \frac{1}{\sqrt{n}}\frac{c\sqrt{a}}{b\sqrt{n}}$$
$$\le \frac{c\sqrt{a\log(n)}}{\sqrt{n}(a-b)}$$

We can say directly that

$$\|\hat{U}_{\cdot 2} \pm U_{\cdot 2}\|_\infty \le \|\hat{U} - UD\|_{2,\infty}$$
$$\implies \quad \text{we need } \frac{c\sqrt{a\log(n)}}{\sqrt{n}(a-b)} \le \frac{1}{\sqrt{n}}$$
$$\frac{a-b}{\sqrt{a}} \ge c\sqrt{\log(n)}$$

Application 2: Limit Theorems Recall on the homework we have the condition that

$$U = ZU_{pure}$$

We might want to test for $i \neq j$ if $Z_{i\cdot} = Z_{j\cdot}$? (e.g. Does Prof Agterberg have the same membership as Prof Yugo?) Say we have a limit theorem for the rows of U, s.t. we can roughly argue that each row is independent of each other. This allows us to run the aforementioned test using the limit theorem we developed (explored in Fan et al. JRSSB "SIMPLE: Statistical Inference on Membership Profiles in Large Networks" 1910.01734) If you are interested in this topic, Prof Agterberg has a StatSci survey paper on limit theorems of this type. For example:

$$A_{ij} \sim \text{Bern}_{i \le j}(P_{ij}) \quad \lambda_k(P) \ge c\log(n)$$
$$\lambda_k(P) \asymp n\rho_n \quad \rho_n \to 0$$
$$P \text{ is rank } K$$
$$(\hat{U} - UO)_i = (EU\Lambda^{-1})_i + o_{\mathbb{P}}\left(\frac{1}{n\sqrt{\rho_n}}\right)$$

Corollary 0.6

$$n\sqrt{P_n}(\hat{U} - UO)_{i\cdot} \xrightarrow{d} \mathcal{N}(0, \Sigma(i))$$
$$n^2\rho_n\Sigma(\cdot) = \lim \operatorname{cov}(EU\Lambda^{-1})$$

Result in $\ell_{2,\infty}$

$$\begin{aligned}
\|\hat{U} - UO\|_{2,\infty} &= \tilde{o}_{\mathbb{P}}\left(\frac{1}{n\sqrt{\rho_n}}\right) \\
\|\hat{U} - UO - EU\Lambda^{-1}\|_{2,\infty} &= o_{\mathbb{P}}\left(\frac{1}{n\sqrt{\rho_n}}\right) \\
\hat{U} = \hat{M}\hat{U}\hat{\Lambda}^{-1} &= (M+E)\hat{U}\hat{\Lambda}^{-1} \\
&= M\hat{U}\hat{\Lambda}^{-1} + E\hat{U}\hat{\Lambda}^{-1} \\
&\approx 0 \approx EU\Lambda^{-1}
\end{aligned}$$

Tang & Priebe, AoS "Limit Theorems for Laplacian" 2016, similar method you can compare. Applications of this are in matrix completion and matrix sensing as well.

Application: Entry-wise Bounds + CIs

Very informally

$$\tilde{M} = \min_{\operatorname{rank}(Z)\le r} \|\hat{M} - Z\|_F = \hat{U}\hat{\Lambda}\hat{U}^\top$$
$$(\tilde{M} - M)_{ij} = \left(U_\perp U_\perp^\top EU\Lambda^{-1}U^\top + U\Lambda^{-1}U^\top EU_\perp U_\perp^\top\right)_{ij} + \text{small order terms}$$

Governments love vertex nomination (esp. in 2020, 2016). Next class, we'll look at optimization algorithms and how we know they'll converge to something good. Turns out the "leave one out" method (spectral initialization) can be adapted for this. We'll monitor the sequences independent of the original noise, this will give us good behavior when they converge.

Lecture Oct 2

Other Non-convex Problems

Definition 0.7

Matrix Factorization

Observe:

$$\hat{M} = M + N \implies M = X_0X_0^\top = \lambda UU^\top$$
$$\hat{X} = \arg\min_{x\in\mathbb{R}^n} \frac{1}{4}\|\hat{M} - XX^\top\|_F^2 \quad \text{this is very well behaved}$$
$$\hat{X} = \hat{\lambda}^{\frac{1}{2}}\hat{U} \quad \hat{U} \text{ is leading eigenvector of } \hat{M}$$

$$\arg\min_{x\in\mathbb{R}^{n\times r}} \frac{1}{4}\|\hat{M} - XX^\top\|_F^2$$

Definition 0.8

Multilayer Linear Neural Networks
Fairly difficult to analyze it (entry-wise nonlinearity), products that add non-convexity to it. This is a next step framework, but theres something else we can do to it.

$$\hat{A}, \hat{B}, ... = \arg\min \frac{2}{L} \|\hat{M} - ABC...\|_F^2$$

$$L = 2 \implies \text{asymmetric MF}$$

More generally:

$$\|\hat{M} - \sigma(A)\sigma(B)...\|_F^2$$

$$\|Y - \sigma(A)\sigma(B)...\|_F^2$$

σ is non linearity

Definition 0.9

Matrix Sensing

$$Y_i = \langle A_i, M \rangle + \varepsilon_i$$

ε is noise
Observe $A_i \sim$ Gaussian$(0, \frac{1}{M^2})$ **GOE**$(\frac{1}{M})$ (Symmetric). M is rank r.
Observe Y_i

$$L^C(M) = \sum_{i=1}^{n} (Y_i - \langle A_i, M \rangle)^2 + \lambda \|M\|_*$$

$$\|M\|_* = \sum_{i=1}^{rank(M)} \sigma_i(M)$$

The smallest convex function that induces a constraint is the nuclear norm.

$$\lambda \sum_{i=1}^{n} \mathbf{1}\{\sigma_i(M) > 0\}$$

L_1 normal smallest convex relaxation of the L_0 norm (somewhat the count of the rank).
Not always differentiable, need to run projective sub gradient descent. Need to run full SVD for each M. Applications are often from MRI machines so M is often a large dim matrix.

$$L^{nc}(X) = \frac{1}{4} \sum_{i=1}^{n} (Y_i - \langle A_i, XX^\top \rangle)^2$$

We need to know the rank, but we might not always know, so we overshoot the rank. This hard-codes the rank constraints, whereas the $*$ norm soft-codes it. When we have the rank right, this is better.

Definition 0.10

Matrix Completion
We like to impute the missing values.

$$\begin{pmatrix} X & 3 & X & 0 \\ 2 & 100 & 4 & X \\ 1 & X & 22 & X \end{pmatrix}$$

(Observations are drawn independently with probability P)

$$\hat{M}_{ij} = M_{ij} + \varepsilon \quad i, j \in \Omega$$
$$L^{nc} = \frac{1}{4P} \sum_{(i,j)\in\Omega)} (\hat{M}_{ij} = (XX^\top)_{ij})^2$$

Definition 0.11

GLMs

$$Y_i = g(X_i, \beta) + \varepsilon_i \quad X_i \in \mathbb{R}^P$$
$$\hat{\beta} = \arg\min_\beta \sum_{i=1}^{n} (Y_i - g(X_i, \beta))^2$$

Over-parametrization occurs when the dimensions of β is greater than samples. (Prof A disagrees with the notion that this is over parametrized, it is correctly parametrized we just don't have sufficiently many observations to match it).

So what is true over-parameterization? $\beta = A \cdot B \cdot C \cdot ...$ now this is overparatermized. β is say a P dimensional object. If these $A, B, C, ...$ have larger inner dimension or same then it will have greater dimension than P. (Not when the model is over-parametrization, but the parameter itself.)

Definition 0.12

Phase retrieval

Non-linear but everything is a vector. We're scaling everything so we only care about it's modulus in some sense and in many cases X can hold complex values.

$$Y_i = a_i^\top M a_i + \varepsilon_i, M = X_0 X_0^\top$$
$$\min_x \frac{1}{4} \sum_{i=1}^{n} (Y_i - (a_i^\top X)^2)^2$$

Definition 0.13

Blind De-convolution

Engineering problem, often complex valued.

$$Y_i = a_i^* X_0 X_0^* b_i + \varepsilon_i$$

Definition 0.14

Tensor Power Iteration / Tensor Completion

Essentially alternating least squares. We observe for tensor low-rank $\mathcal{M}$ (this can mean many things, see C-P rank, Tucker rank, tensor training rank)

$$\hat{\mathcal{M}} = \mathcal{M} + \mathcal{N}$$

Why do we care?

1. We want to know the tolerance of the noise dependence (for rates, inference, computational, properties of local minima purposes)

2. To study the behavior of critical points (why? if we need to run gradient descent we'd like to know this behavior)

3. We'd like to study the optimization landscape (if we understand the landscape than we can understand 1 and 2)

Statistically, we understand convex conditions due to the nice KKT behaviors, e.g. convex LASSO is well understood. Note LASSO as opposed to RIDGE does not have a closed form solution.

1) Matrix Factorization

We'd like to understand gradient descent itself.

Lemma 0.15

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ satisfies $\nabla^2 f(x) \succcurlyeq \alpha I_n$ (strong convexity), $\|\nabla^2 f(x)\| \preccurlyeq \beta$ (β-smooth)
Set X_{opt} as any local min of f.

$$X_{opt} \in B_\xi = \{x : \|X - X_{opt}\| \leq \xi$$

$$X_{t+1} = X_t - \eta \nabla f(x_t) \quad \text{let } \eta = \frac{1}{\beta}$$

Suppose $x \in B_\xi$ then

$$\|X_t - X_{opt}\| \leq (1 - \frac{\alpha}{\beta})^t \|X_n - X_{opt}\|$$

Proof (sketch)

We know that $\nabla f(X_{opt}) = 0$

$$\begin{aligned}
X_{t+1} - X_{opt} &= X_t - \eta \nabla f(X_t) - (X_{opt} - \eta \nabla f(X_{opt})) \\
&= X_t - X_{opt} - \eta \underbrace{(\nabla f(X_t) - \nabla f(X_{opt}))}_{\Gamma} \\
&= X_t - X_{opt} - \eta \left[\int_0^1 \nabla^2 f(tX_t + (1-t)X_{opt}) dt \right] (X_t - X_{opt}) \\
&= \left[I - \eta \int \nabla^2 ... \right] (X_t - X_{opt}) \\
&\implies \|X_{t+1} - X_0\| \leq \|I - \eta\Gamma\| \|X_t - X_0\| \\
0 \leq \lambda(I - \eta\Gamma) &\leq (1 - \eta\alpha) = (1 - \frac{\alpha}{\beta})
\end{aligned}$$

Provided $tX_t t(1-t)X_{opt} \in B_\xi$ true if $X_t \in B_\xi$ apply induction.

Claim 0.16

$$\begin{aligned}
M = \tilde{X}\tilde{X}^\top &= \lambda U U^T \\
f(x) &= \frac{1}{4} \|M - XX^T\|_F^2 \\
\nabla f(x) &= -(M - XX^\top)X \\
\nabla^2 f(x) &= \|x\|^2 I + 2XX^\top - M
\end{aligned}$$

Theorem 0.17

If $\|X_0 - \tilde{X}\| \leq \frac{1}{15\sqrt{\lambda}}$ then $\eta = \frac{1}{4.5\lambda}$

$$\|X_t - \tilde{X}\| \leq \left(1 - \frac{1}{18\sqrt{\lambda}}\right)^t \|X_0 - \tilde{X}\|$$

Nice behavior while non-convex, restricted strong convexity. From HW 1:

$$\nabla^2 f(x) \succcurlyeq C' I_n, C' = \frac{1}{4}\lambda \implies \|X_t - X_0\| \leq \|I - \eta\Gamma\|\|X_t - X_0\|$$

$$0 \leq \lambda_i(I - 3\Gamma) \leq (1 - 3\alpha) = (1 - \frac{\alpha}{\beta})$$

$$\|\nabla^2 f(x)\| \leq 4.5\lambda$$

Apply Lemma

2) Matrix Sensing

$$\mathcal{A} : \mathbb{R}^{m\times m} \longmapsto \mathbb{R}^n, \quad \mathcal{A}(M) = \begin{pmatrix} \langle \mathcal{A}_1, M\rangle \\ ... \\ \langle \mathcal{A}_n, M\rangle \end{pmatrix}$$

$$\begin{aligned} f(x) &= \frac{1}{4}\sum_{i=1}^{n}(Y_i - \langle A_i, XX^\top\rangle)^2 \\ &= \frac{1}{4}\|Y - \mathcal{A}(XX^\top)\|^2 \\ &= \frac{1}{4}\|\mathcal{A}(M) + \varepsilon - \mathcal{A}(XX^\top)\|^2 \end{aligned}$$

Special Case $\sum = 0$, $\mathcal{A}^*(Y) = \sum_{i=1}^n Y_i A_i$

$$\frac{1}{4}\|\mathcal{A}(M - XX^\top)\|^2$$

$$\begin{aligned} \nabla f(x) &= \mathcal{A}^*\mathcal{A}(XX^\top - M)X \\ \mathbb{E}_{\mathcal{A}}\nabla f(x) &= (XX^\top - M)X \end{aligned}$$

Restricted isometry property guarantees that

$$\mathcal{A}^*\mathcal{A} \approx I$$

Overall rank $\leq 2r$ matrices